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DIAGNOSTIC VALUE OF PLASMA 1,5-ANHYDROGLUCITOL FOR DIAGNOSING PREDIABETES AMONG OBESE YOUNG ADULTS

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ABSTRACT

Backgrounds: Prediabetes is characterized by elevated blood glucose levels that do not yet meet the criteria for diabetes. Early detection is crucial, particularly in obese young adults who are at higher risk. Traditional diagnostic methods, such as fasting blood sugar (FBS), oral glucose tolerance test (OGTT), and Hemoglobin A1c (HbA1c), have limitations. This study aims to assess the diagnostic value of plasma 1,5-Anhydroglucitol (1,5-AG) for detecting prediabetes in this population.
Methods: This diagnostic study was conducted over six months at Dr. M. Djamil General Hospital, involving 143 participants selected through incidental sampling. Plasma 1,5-AG and HbA1c levels were measured. Diagnostic accuracy was evaluated using a 2x2 table, with HbA1c as the gold standard.
Results: Of the 143 participants, 55.2% were male, with an average age of 30.4 years and mean BMI of 29.97. A family history of diabetes was reported by 30.8% of participants. Plasma 1,5-AG testing demonstrated a sensitivity of 90.79%, specificity of 74.79%, positive predictive value of 36.73%, and negative predictive value of 97.87%.
Conclusions: Plasma 1,5-Anhydroglucitol shows potential as an alternative screening tool for prediabetes, offering high sensitivity and a strong negative predictive value.

KEYWORDS: Prediabetic state, obesity, young adult, 1,5-Anhydroglucitol

INTRODUCTION

Prediabetes is a condition where blood sugar levels are elevated but not yet high enough for a diabetes diagnosis. First introduced by the Department of Health and Human Services (DHHS) and the American Diabetes Association (ADA) in 2002, prediabetes affects millions worldwide, with 314 million cases in 2019 and a projected rise to 418 million by 2025. Without intervention, 4–9% of prediabetes cases progress to diabetes each year.[1,2]
In the U.S., the CDC estimates that one in three adults may have prediabetes, many of whom are unaware of their condition. Indonesia has also seen rising prediabetes rates, reaching 30.8% in 2018. Early detection is essential, especially for young adults, as early onset increases the risk of long-term complications.[3–6]
Current diagnostic tests, including fasting blood sugar (FBS), oral glucose tolerance test (OGTT), and Hemoglobin A1c (HbA1c), possess several limitations. FBS and OGTT can be influenced by temporary factors such as stress or illness, leading to variability in results.[2,7] Moreover, HbA1c can be unreliable in specific clinical situations, such as

anemia, kidney disease, and hemoglobinopathies. These limitations underscore the urgent need for alternative diagnostic approaches that provide accurate assessments of glycemic control.[1,7]

Biomarkers like 1,5-Anhydroglucitol (1,5-AG) offer a promising solution. 1,5-AG reflects short-term glycemic fluctuations, providing a more dynamic view of glucose control compared to HbA1c. Its stability and affordability make it an attractive option for early diagnosis of prediabetes.[8–10] This study aims to evaluate the diagnostic value of plasma 1,5-AG in detecting prediabetes among young adults with obesity, addressing gaps in current diagnostic approaches.

METHODS

Study Design and Setting

This diagnostic study was conducted at RSUP DR. M. Djamil Padang over a six-month period. The study population consisted of individuals with obesity aged 18 to 40 years, and a total sample size of 143 was determined based on specific inclusion and exclusion criteria.

PARTICIPANTS

Inclusion criteria included individuals with a BMI greater than 25, aged between 18 and 40 years, and who were willing to participate. Exclusion criteria encompassed a history of prediabetes or diabetes mellitus, kidney disorders, the use of SGLT2 inhibitors or corticosteroids, and erythrocyte abnormalities. Potential participants were selected through incidental sampling, with baseline data such as name, age, sex, family history of diabetes, waist circumference, weight, height, and BMI collected at the Internal Medicine Poly Clinic.

DATA COLLECTION AND ANALYSIS

Blood samples of 6 mL were taken for testing HbA1c and 1,5-Anhydroglucitol. Descriptive analysis was performed on patient characteristics, while sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) for the 1,5-Anhydroglucitol test were calculated in comparison to HbA1c using a 2x2 table.

ETHICAL CONSIDERATIONS

Ethical clearance was obtained from the Health Research Ethics Committee of RSUP Dr. M. Djamil Padang prior to the study, under ethical approval number **DP.04.03/D.XVI.XI/622/2023**. Participants were provided with comprehensive information regarding the research. Written consent was secured, and participation was voluntary, with all associated costs covered by the researchers.

RESULT

BASELINE CHARACTERISTICS OF THE STUDY POPULATION

The baseline characteristics of young adults with obesity are summarized in Table 1. Key demographic and clinical variables include age, gender, body mass index (BMI), waist circumference, family history of diabetes, and occupation. These characteristics provide an overview of the study population, serving as essential factors in assessing the diagnostic performance of 1,5-Anhydroglucitol (1,5-AG) for prediabetes detection in obese individuals.

Table 1. Baseline Characteristics of the Study Population

Characteristics	n (%)	Mean (SD)	p
Gender			0.656
Male	79 (55.2)		
Female	64 (44.8)		
Age (years)		30.4 (3.88)	0.007
18-24 years	12 (8.39)		
25-29 years	29 (20.28)		
30-34 years	86 (60.14)		

35-40 years	16 (11.19)		
Occupation			0.021
Resident Doctor (PPDS)	129 (90.21)		
General Practitioner	14 (9.79)		
Body Mass Index (kg/m²)		29.97 (3.49)	0.949
Obesity I (25-29.9)	83 (58.05)		
Obesity II (30-34.9)	51 (35.66)		
Obesity III (≥35)	9 (6.29)		
Family History of Diabetes			0.033
Diabetes	44 (30.8)		
No diabetes	99 (69.2)		
Waist Circumference			0.615
Male	94.47 (10.42)		
>90cm	73 (92.41)		
≤90cm	6 (7.59)		
Female	99.97 (9.55)		
>80cm	62 (96.88)		
≤80cm	2 (3.12)		
Hemoglobin A1c (HbA1c)		5.08 (0.66)	
1,5-Anhydroglucitol		7.89 (4.76)	

Hemoglobin A1c (HbA1c) and 1,5-Anhydroglucitol Results

HbA1c levels were measured in all study participants. The frequency distribution of prediabetes in young adults with obesity, based on HbA1c results, is presented in Table 2. The frequency distribution of prediabetes in young adults with obesity based on 1,5-AG levels can be seen in Table 3.

Table 2. Frequency Distribution of Prediabetes in Young Adults with Obesity Based on Hemoglobin A1c (HbA1c) Levels

HbA1c	**n**	**%**
Non-Prediabetes	123	86
Prediabetes	20	14
Total	143	100

Cut-off point: prediabetes: 5.7–6.4%, diabetes > 6.4%
*Five samples fell into the diabetes category.

Table 3. Frequency Distribution of Prediabetes in Young Adults with Obesity Based on Plasma 1,5-Anhydroglucitol

1,5-AG	**n**	**%**
Non-Prediabetes*	94	65.7
Prediabetes	49	34.3
Total	143	100

Diagnostic Value of 1,5-Anhydroglucitol in Diagnosing Prediabetes in Young Adults with Obesity

A total of 143 young adults with obesity were tested for suspected prediabetes. All samples were assessed using the HbA1c method, standardized by the National Glycohemoglobin Standardization Program (NGSP). Based on HbA1c results, 20 individuals were diagnosed with prediabetes. The sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) of plasma 1,5-AG compared to HbA1c were calculated using a 2x2 table, as shown in Table 4 below.

Table 4. Diagnostic Test for Plasma 1,5-Anhydroglucitol in Young Adults with Obesity

HbA1c	**Prediabetes**	**Non-Prediabetes**	**Total (TP+FP)**
1,5-Anhydroglucitol			
Prediabetes	18 (TP)	31 (FP)	49 (TP+FP)
Non-Prediabetes	2 (FN)	92 (TN)	94 (FN+TN)
Total	20 (TP+FN)	123 (FP+TN)	143 (n)

*TP *(True Positive)*; FP *(False Positive)*; FN *(False Negative)*; TN *(True Negative)*

Sensitivity	: TP/(TP+FN)	= 18/(18+2) x 100%	= 90%
Spesifiсity	: TN/(FP+TN)	= 92 /(31+92) x 100%	= 74,79%
Positive Predictive Value	: TP/(TP+FP)	= 18/(18+31) x 100%	= 36,73%
Negative Predictive Value	: TN/(FN+TN)	= 92/(2+92) x 100%	= 97,87%

DISCUSSION

In this study, the proportion of young adult males with obesity was higher than that of females, with 55.2% males compared to 44.8% females. These results align with previous studies, such as Kitchlew et al. (2017) in Pakistan, which reported higher obesity prevalence among males aged 18-30, where 54.7% of male participants were categorized as obese.[11] A similar study by Peltzer et al. (2014), involving 15,746 university students from 22 developing countries, found that 22% of young adults were obese, with a higher prevalence in men (24.7%) compared to women (19.3%).[12] However, data from the National Health and Nutrition Examination Survey (NHANES) in 2017-2018 indicated no significant difference in obesity prevalence between men and women aged 20-40. Variations in obesity distribution by gender in different studies may be attributed to demographic factors, dietary habits, and other contributing factors.[13] This difference can also be explained by body composition between genders. Men generally have a higher lean body mass than women, while women tend to store fat peripherally, especially in the legs and hips. In contrast, men are more prone to higher visceral fat, which is associated with negative metabolic outcomes such as increased insulin levels, free fatty acids, and postprandial triglycerides. Higher visceral fat in men is also more strongly linked to obesity-related complications compared to peripheral fat distribution in women, which is often associated with lower health risks.[14,15]

The largest age group in this study was 30-34 years (60.14%), with participants ranging from 22 to 37 years old and a mean age of 30.4 years. Kitchlew et al. (2017) also reported a mean age of 32.0 years in their study on the prevalence of prediabetes in obese adults in Pakistan, while Decroli et al. (2024) noted a mean age of 31.91 years in their study on the correlation between Body Mass Index (BMI) and testosterone levels in obese individuals with prediabetes. This study also suggests that the risk of prediabetes increases with age, with prediabetes often appearing earlier in younger individuals. A decline in physical activity during adolescence may also contribute to obesity development in adulthood.[11,16] Guthold et al. (2020) reported that the majority of adolescents worldwide (approximately 81%) do not meet the recommended daily physical activity levels, and this sedentary behavior can exacerbate the risk of adult obesity. This is further supported by the finding that the majority of the sample in this study (90.21%) comprised specialist students, with 9.79% being young doctors, indicating a significant age-related influence on prediabetes risk.[17–19]

Moreover, obesity is not only associated with an increase in overall body weight but also with fat distribution that affects insulin resistance risk. Obese men have higher visceral fat tissue, which is known to increase the risk of metabolic complications, such as diabetes mellitus. A study by Adamo et al. (2003) showed that obese adolescents with impaired glucose tolerance (IGT) are more prone to insulin resistance than adolescents with normal glucose tolerance, despite having similar fat mass.[18,19] The study revealed that the mean BMI of the sample was 29.97, with the largest group being individuals with a BMI of 25-29.9 (58.05%). BMI is commonly used to assess obesity, although it has limitations because it does not differentiate between fat mass and lean mass. A study by Bacong et al. (2024) in California and Hawaii found that the distribution of obesity was higher in individuals with class I obesity than in those with class II and III obesity. Meanwhile, Andes et al. (2020) reported a mean BMI of 27.7 in their study on prediabetes among adolescents and young adults. Although BMI is often used as an obesity indicator, research shows that BMI alone is insufficient for accurately identifying fat distribution.[5,20]

More specific fat distribution can be measured by waist circumference, which is better at predicting obesity-related health risks than BMI. Zhen et al. (2022) reported a mean waist circumference of 89.9 cm in individuals with central obesity, while this study found a mean waist circumference of 94.47 cm. Waist circumference measurements can identify individuals with higher cardiometabolic risk compared to those identified solely by BMI.[21]

In this study, 30.8% of the sample had a family history of diabetes, and the P-value (0.033) indicates that family history has a significant influence on the occurrence of prediabetes. A study by Al Sharaafi et al. (2021) in Yemen also found that obese individuals with close relatives suffering from diabetes have a much higher risk of developing prediabetes and diabetes, with a prevalence of 47.6% among overweight and obese individuals. Similar results were reported by Ramenzakhani et al. (2022) in Iran, where individuals with a family history of diabetes had a 54% higher risk of developing diabetes.[22,23] Family history of diabetes has been proven to be a strong risk factor for prediabetes and diabetes, largely due to shared genetic and environmental factors. Population studies in the United States have shown that the risk of developing diabetes increases with the number of family members affected by the disease.[24,25]

The study found that positive HbA1c results for prediabetes were seen in 14% of participants, while 3.5% were diagnosed with diabetes, and 82.5% were negative. Compared to other studies, such as Kusuma et al. (2023) in Indonesia, which reported 24.86% prediabetes and 7.85% diabetes, and Park et al. (2020) in South Korea, which identified 21.2% prediabetes and 1.4% diabetes in individuals aged 20-40, the prevalence in this study appears lower. The variation in results could be due to differences in age ranges, diagnostic methods, and socio-demographic factors.[26,27]

In this study, 12.58% of young obese adults with 1,5-AG values between 4.5 and 5.5 µg/ml were confirmed to have prediabetes by HbA1c. This is similar to Ohigashi et al. (2022) in Japan, which found 11.64% prediabetes using TTGO, HbA1c, and 1,5-AG. Malkan et al. (2020) in Turkey found that 16.41% of 128 individuals had prediabetes with an average HbA1c of 6.2% and 1,5-AG of 5.6 µg/ml.[28,29]

Amser et al. (2023) reported that obese individuals with prediabetes had significantly higher HbA1c levels than normal-weight groups, but there was no significant difference between obese and normal-weight individuals without prediabetes. The study also showed that 1,5-AG levels were significantly lower in obese individuals with prediabetes, attributed to the competition between glucose and 1,5-AG for reabsorption in the renal tubules. During hyperglycemia, glucose is preferentially reabsorbed, leading to a decrease in 1,5-AG reabsorption and plasma levels.[30]

The diagnostic value of plasma 1,5-AG for detecting prediabetes in young obese adults was demonstrated with a sensitivity of 90%, specificity of 74.79%, positive predictive value of 36.73%, and negative predictive value of 97.87%. A higher sensitivity indicates that 1,5-AG can effectively detect prediabetes, making it useful for screening. Comparatively, Amser et al. (2023) reported a sensitivity of 72.5% and specificity of 85%, while Malkan et al. (2018) reported a sensitivity of 63.3% and specificity of 65.8% with a cut-off value of 5.5 µg/ml for prediabetes.[29,30] Although this study showed a high sensitivity for 1,5-AG, specificity was lower, indicating it is more effective for ruling in prediabetes than ruling it out. After excluding diabetic cases, recalculations yielded a sensitivity of 100% and specificity of 76.27%. However, the positive predictive value remained low, likely due to the small number of prediabetes cases identified.

The study observed 31 false positives with 1,5-AG levels between 4.5-5.5 µg/ml, suggesting the need for regular monitoring in these individuals. Chronic hyperglycemia leads to beta-cell dysfunction and insulin resistance. Studies by Xu et al. (2024) found that reduced 1,5-AG levels were associated with decreased insulin sensitivity and beta-cell function in newly diagnosed diabetes patients.[31] Similarly, Jiménez-Sánchez et al. (2022) noted that serum 1,5-AG strongly correlates with pancreatic beta-cell mass, highlighting its potential for monitoring beta-cell function before the onset of diabetes.[32] However, this study has limitations, as it did not include additional assessments, such as bioimpedance analysis, dual-energy X-ray absorptiometry, or other tests to evaluate obesity in the population and sample.

CONCLUSION

The study concluded that the sensitivity of the 1,5-Anhydroglucitol (1,5-AG) plasma test for diagnosing prediabetes in young adults with obesity was 90.79%, while its specificity was 74.79%. The positive predictive value of the 1,5-AG plasma test for diagnosing prediabetes in this population was 36.73%, and the negative predictive value was notably high at 97.87%. The prevalence of prediabetes detected through the 1,5-AG plasma test was 34.3%, while the prevalence of prediabetes determined by Hemoglobin A1c (HbA1c) testing was lower, at 14%. These findings suggest that the 1,5-AG plasma test may serve as a highly sensitive tool for early detection of prediabetes in young adults with obesity.

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